HW3P2 Bootcamp

Utterance to Phoneme Mapping using Sequence Models

(Spring 2026)

Reflection of HW1P2 & HW2P2

1. What was challenging but you figured that out?

2. Which strategies / resources helped you the most?

3. What would you do differently if you could start over?

Reflection of HW1P2 & HW2P2

1. What was challenging but you figured that out? → Great job
2. Which strategies / resources helped you the most? → Things to keep for HW2P2
3. What would you do differently if you could start over? → Things to change for HW2P2

Logistics

- Early submission is due **March 20, 11:59PM ET**
  - Kaggle submission a with Lev. Dist <= 12
  - Canvas MCQ
- On time submission deadline: **March 27,2025, 11:59 PM ET**
- Constraints: No attention
- Parameter limit: 20M

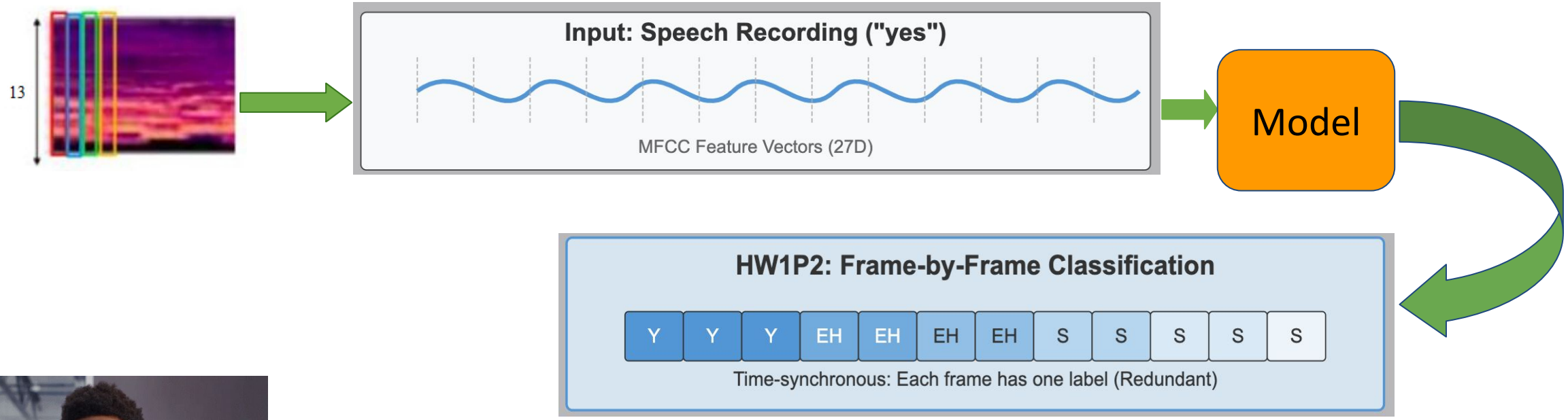
Notice

- This homework requires more time ….
  - to complete the notebook
  - to train the network: sometimes >100 epoches
- Models must be written yourself and trained from scratch
- Please start early!
- Use what you learned from HW1P2 💪
- Strategize with your Study Team.

Agenda

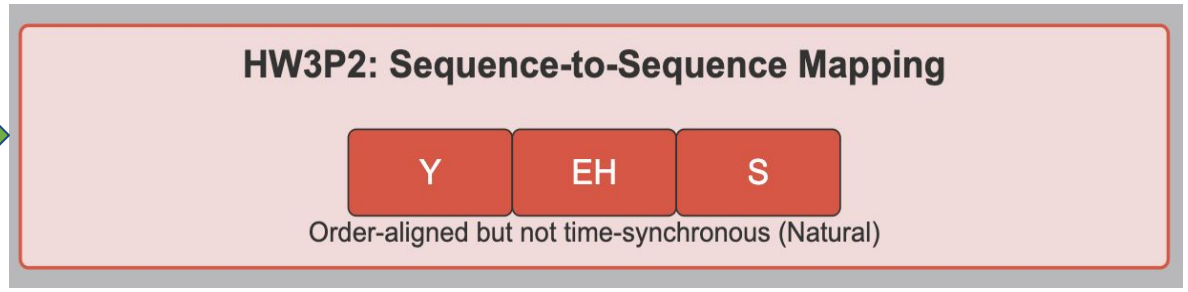
1. HW1P2 Recap
2. HW3P2 Overview
3. ASR Model Setup
   a. Sequence Model Overview
   b. Embedding Layer
   c. Encoder
   d. Decoder
4. Hyperparameters and Regularization
5. Final Tips

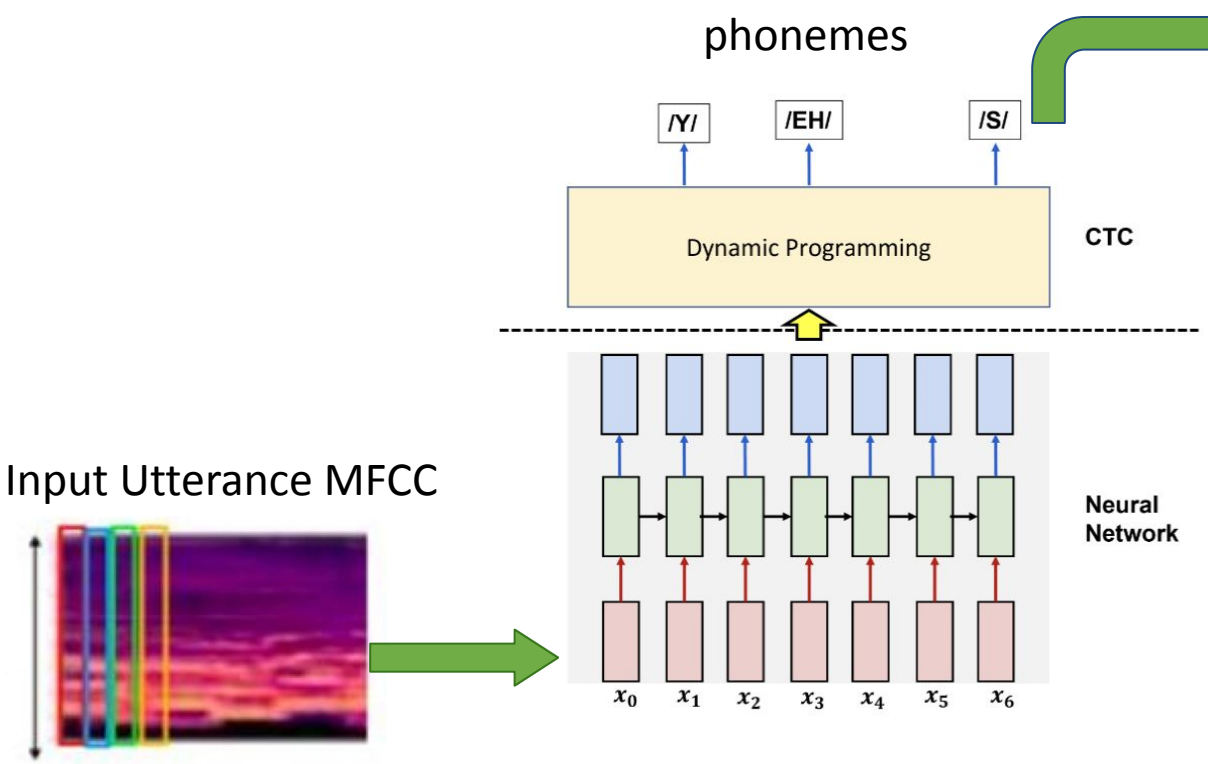
Hw1p2 Frame-Frame Classification



What is the problem with this 😥?

- Redundant phonemes predictions
- Not Natural (Non Human)
- Time Synchronous.



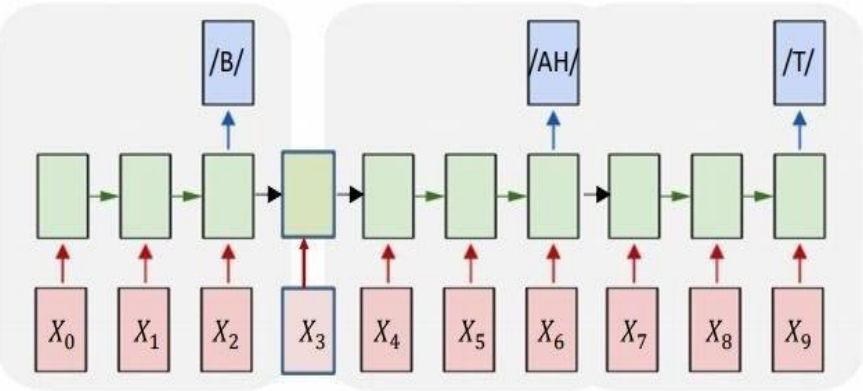


HW1 ⟹ "/Y/ /Y/ /Y/ ... /EH/ /EH/ ... /S/ /S/..." for a simple word like "yes".

HW3 ⟹ "/Y/ /EH/ /S/ "

How do we go about this🤨?

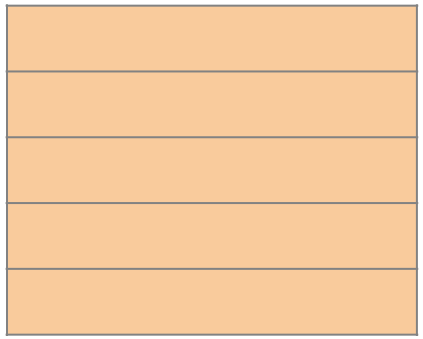
- Features: Same as HW1P2 (27D)
- Labels: Order synchronous but not time synchronous
- Should output sequence of phonemes
  - [‘B’, ‘IH’, ‘K’, ‘SH’, ‘A’] (precisely the indexes)
- Loss: CTCLoss([Pytorch](#) 😍)
- Metric: mean Levenshtein distance
  - Provided in a nice clean looking starter notebook
  - Sequence of Phonemes -> String and then calculate distance (Use CMUdict and ARPABet)



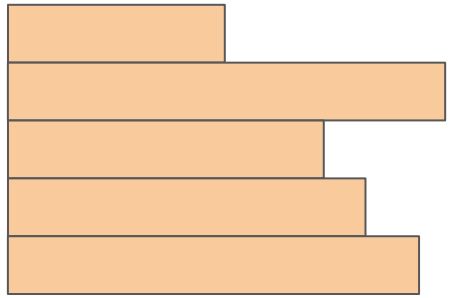
Happy 🤗 …….?

Batch of Variable Length Inputs 🧐

Equal length Input



variable length Input

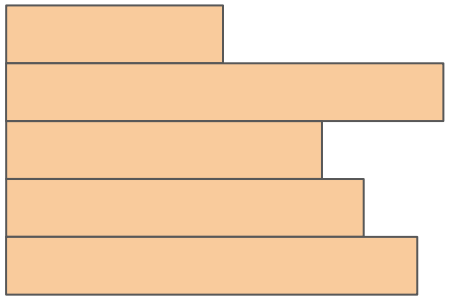


- HW1, HW2: Equal length inputs
- HW3: Variable Length sequences

How do we handle this type of input❗⁉

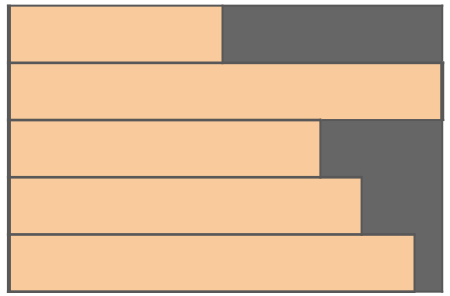
Batch of Variable Length Inputs: 🤲Padding

Padding



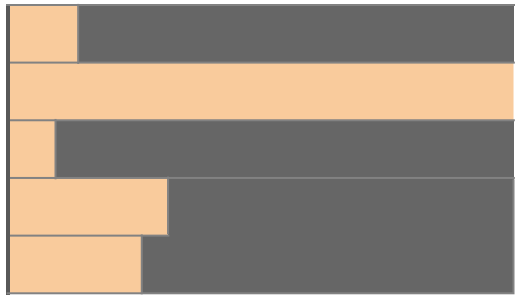
Need to store unpadded lengths as well.
Have the variables *lengths_x, lengths_y* in the starter notebook

Padded to equal lengths



(B, *, 27) → (B, T, 27)

Problematic Example (When padding on whole dataset)



Inefficient with space

Ref: 11785 Fall 24 Bootcamp

Batch of Variable Length Inputs: Packing

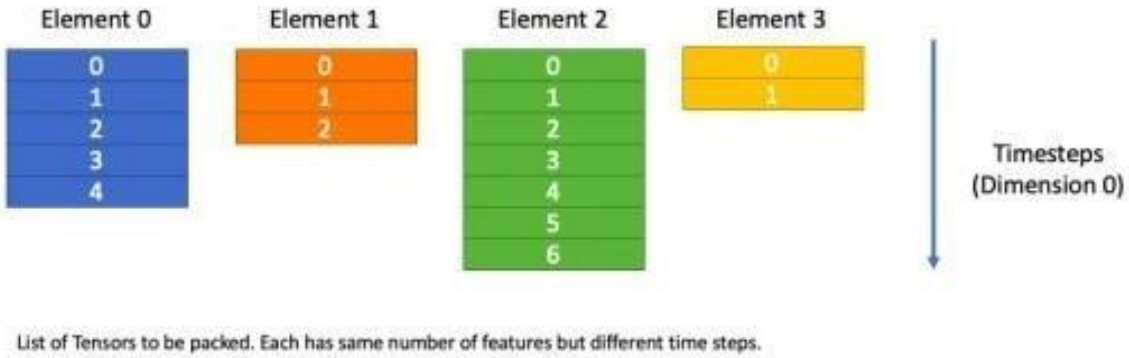


Figure 2: List of tensors we want to pack

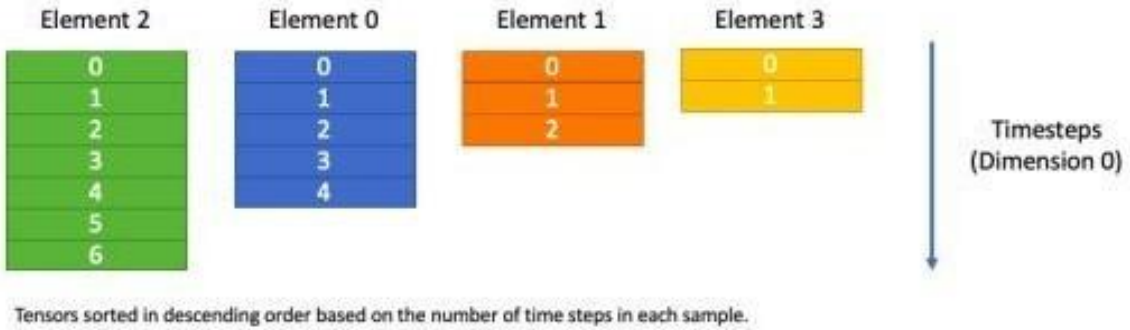
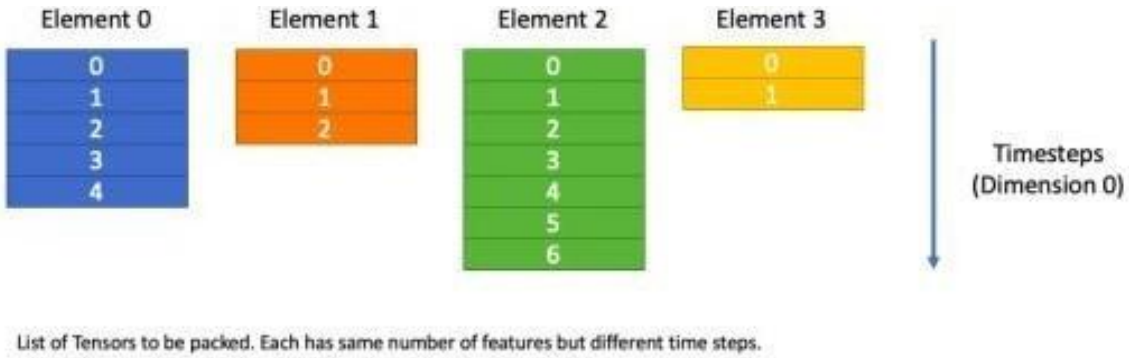


Figure 3: First we sort the list in a descending order based on number of timesteps in each

Ref: 11785 Fall 24 Bootcamp

Batch of Variable Length Inputs: Packing



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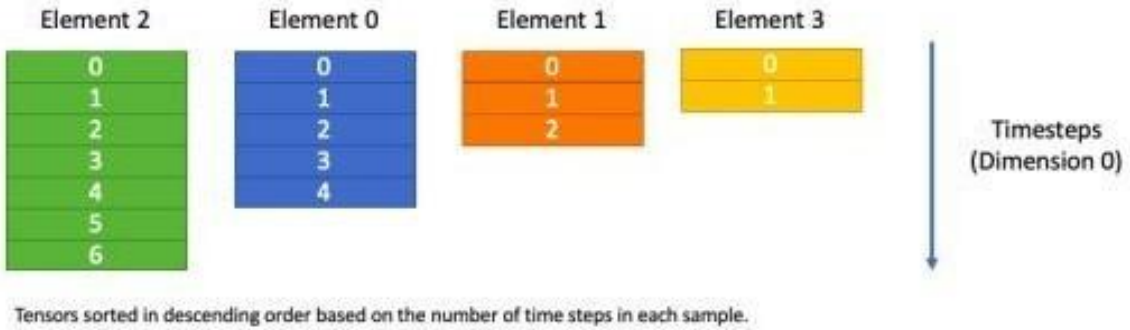


Figure 3: First we sort the list in a descending order based on number of timesteps in each

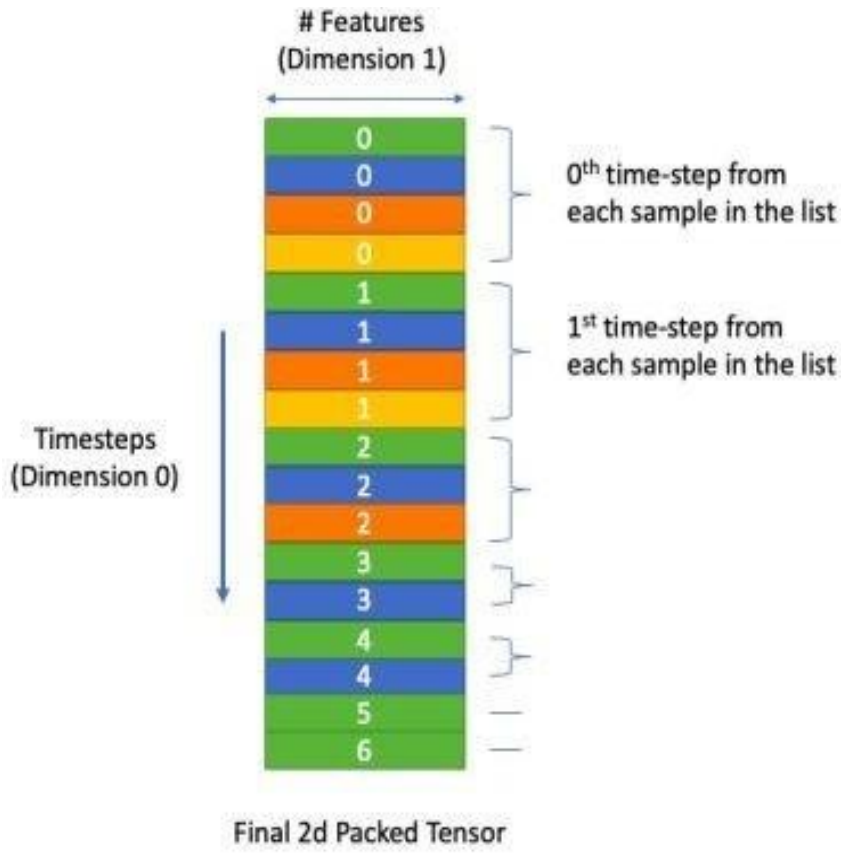


Figure 4: Final Packed 2d Tensor

Ref: 11785 Fall 24 Bootcamp

Packed Sequence

- Pad_sequence()
  - Pads to equal length for batching
- pack_padded_sequence()
  - Packs batch of padded sequences
  - Requires sequences + sequence lengths
- X = pad_packed_sequence()
  - Unpacks back to a batch of padded sequences
  - Outputs sequences + sequence lengths
- Collate Function
  - Dataloader argument
  - Helpful when altering data for batch

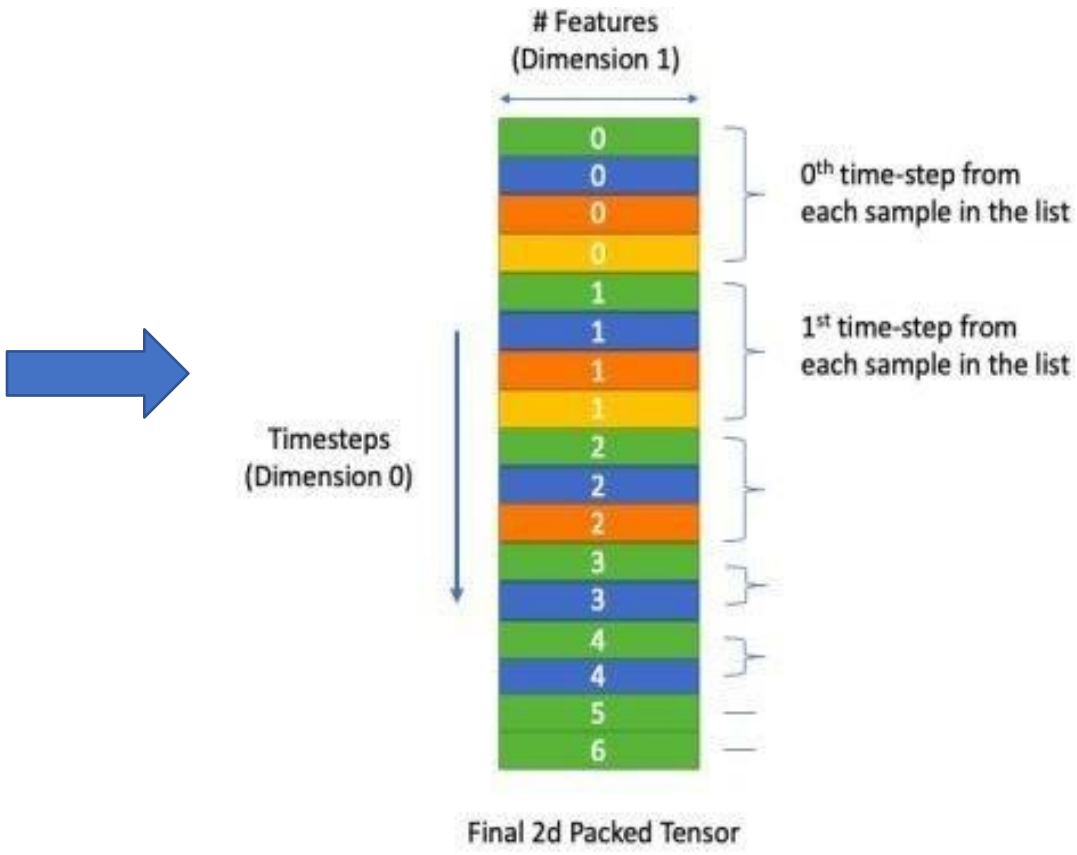
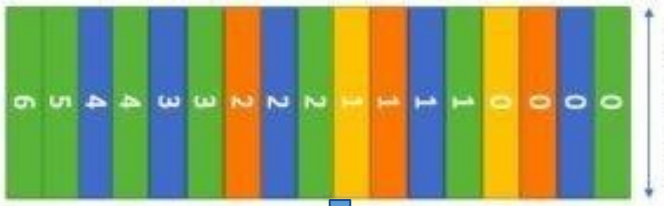
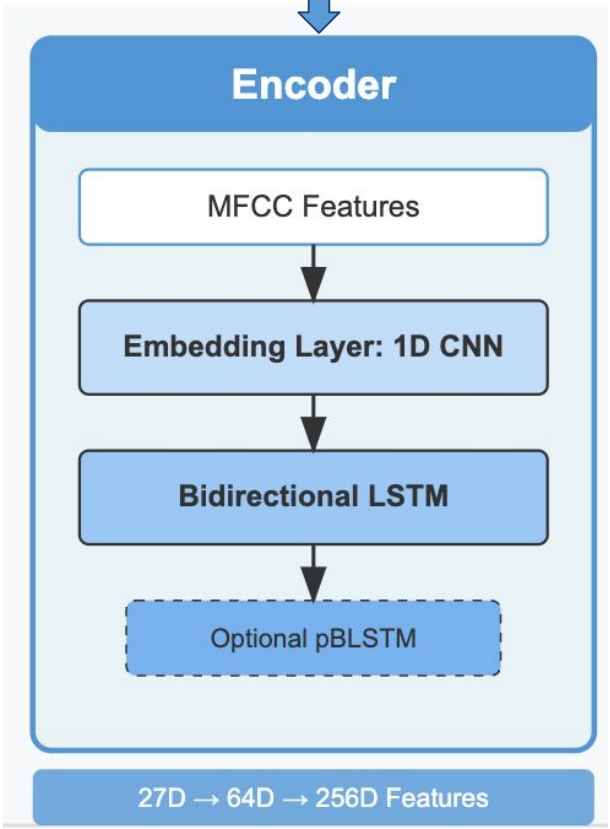
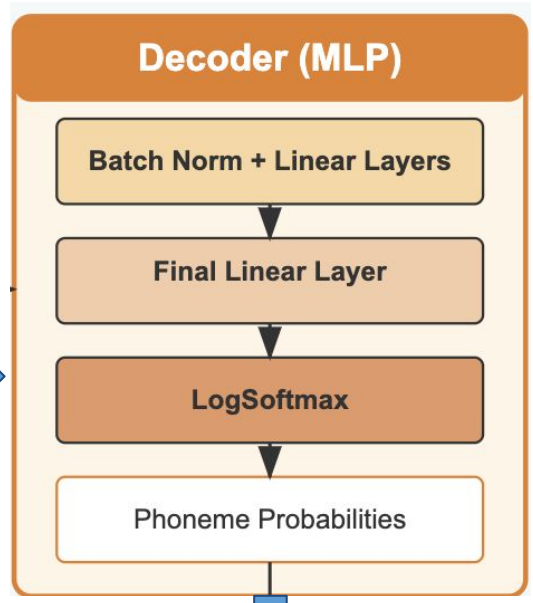


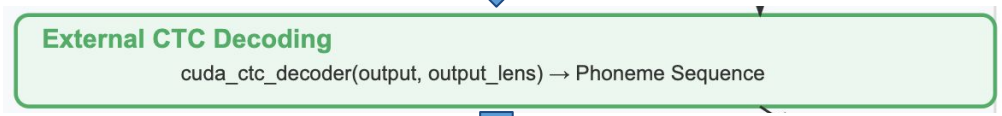
Figure 4: Final Packed 2d Tensor

ASR Model Setup: Encoder - Decoder









/Y/ /EH/ /S/

Encoder

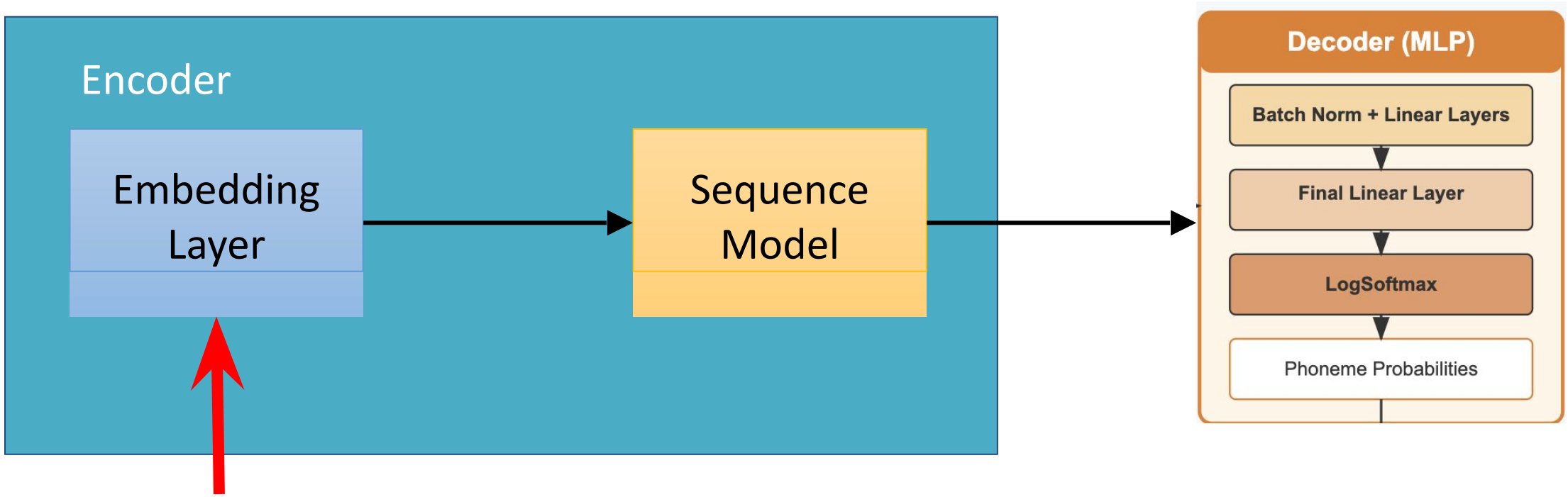
- Typically used to generate high-level representations of given input data.
- There are no labels used to train encoders
- Are trained jointly with decoders.
- Can be any network, CNN, RNN or Linear

Decoder

- It is a network that takes in the feature representation from the Encoder and tries to generate the closest match to the expected output.
- Loss function is applied on the output of the Decoder.
- Can also be trained without encoders, encoders are basically to amplify the results of the decoder

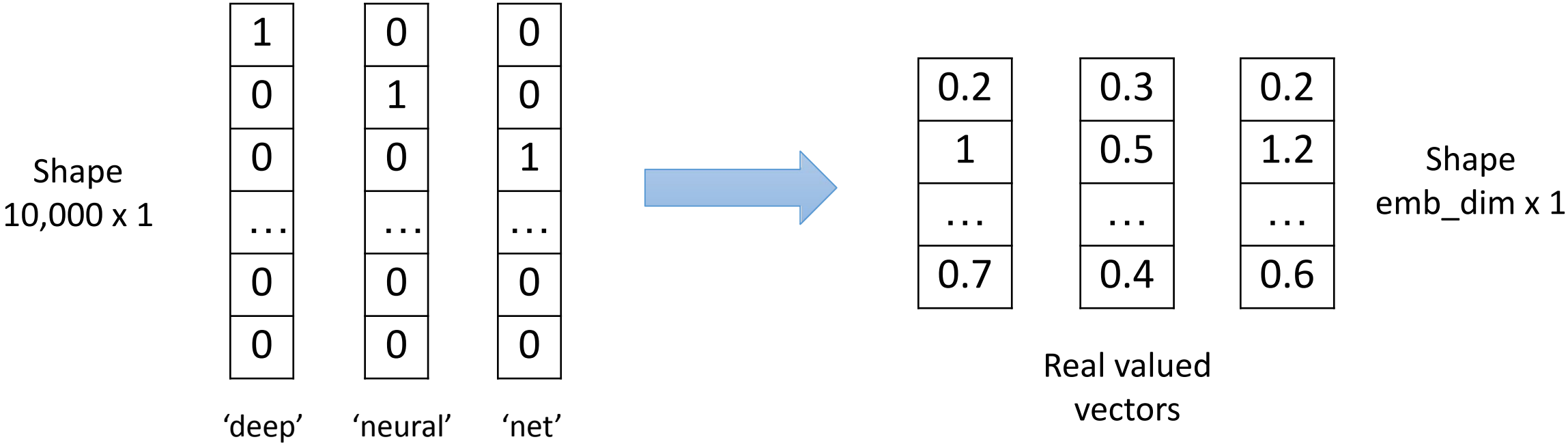
Embedding Layer

- Optional but recommended
- Used to increase/decrease the dimensionality of the input



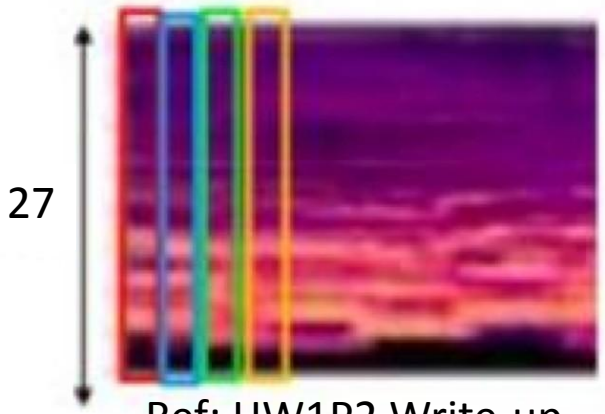
Embedding Layer

- Optional but recommended
- Used to increase/decrease the dimensionality of the input
- Eg. In NLP, 10k vocabulary represented as 1 hot vectors with 10k dim



Embedding Layer

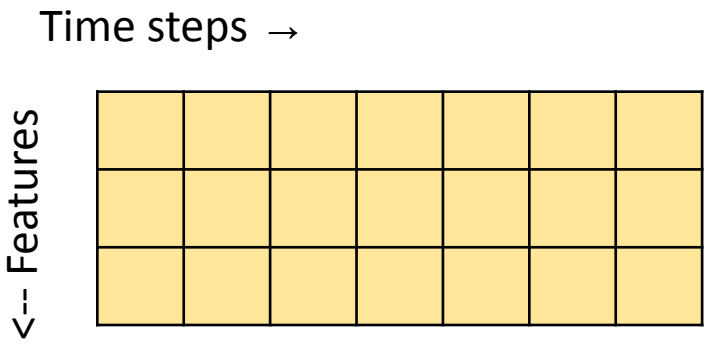
- Optional but recommended
- Used to increase/decrease the dimensionality of the input
- Our task:
  - Input dim = 27
  - Expand to emb_dim > 27 for feature extraction



Ref: HW1P2 Write-up

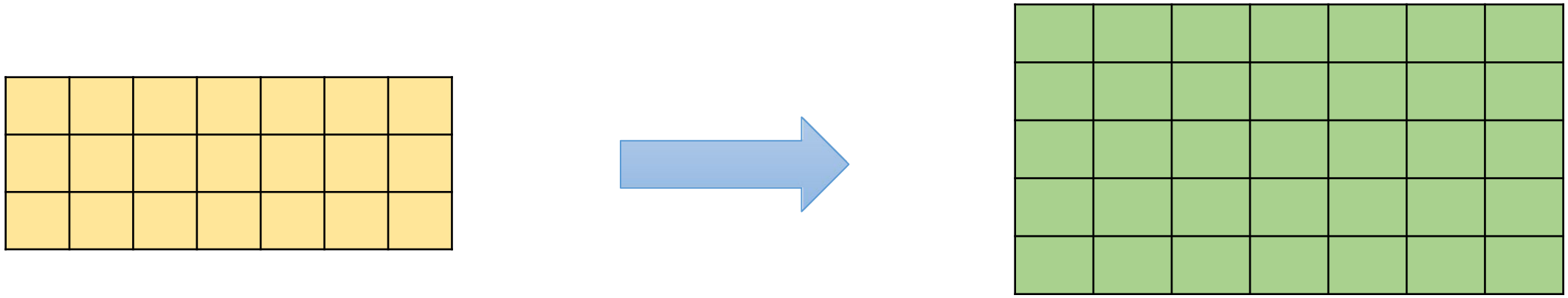
Embedding Layer: Conv1d Layers

- Consider the below as an input having 3 features at each time instant



Embedding Layer: Conv1d Layers

- We can use Convolution to which increases the channels of the input as we go deeper.



- No. Filters = 5
- Kernel= 3; Padding= 1; Stride= 1 **3D**
- Kernel= 5; Padding= 2; Stride= 1 **5D**
  (Or anything similar)

Embedding Layer: Conv1d Layers

Objective:

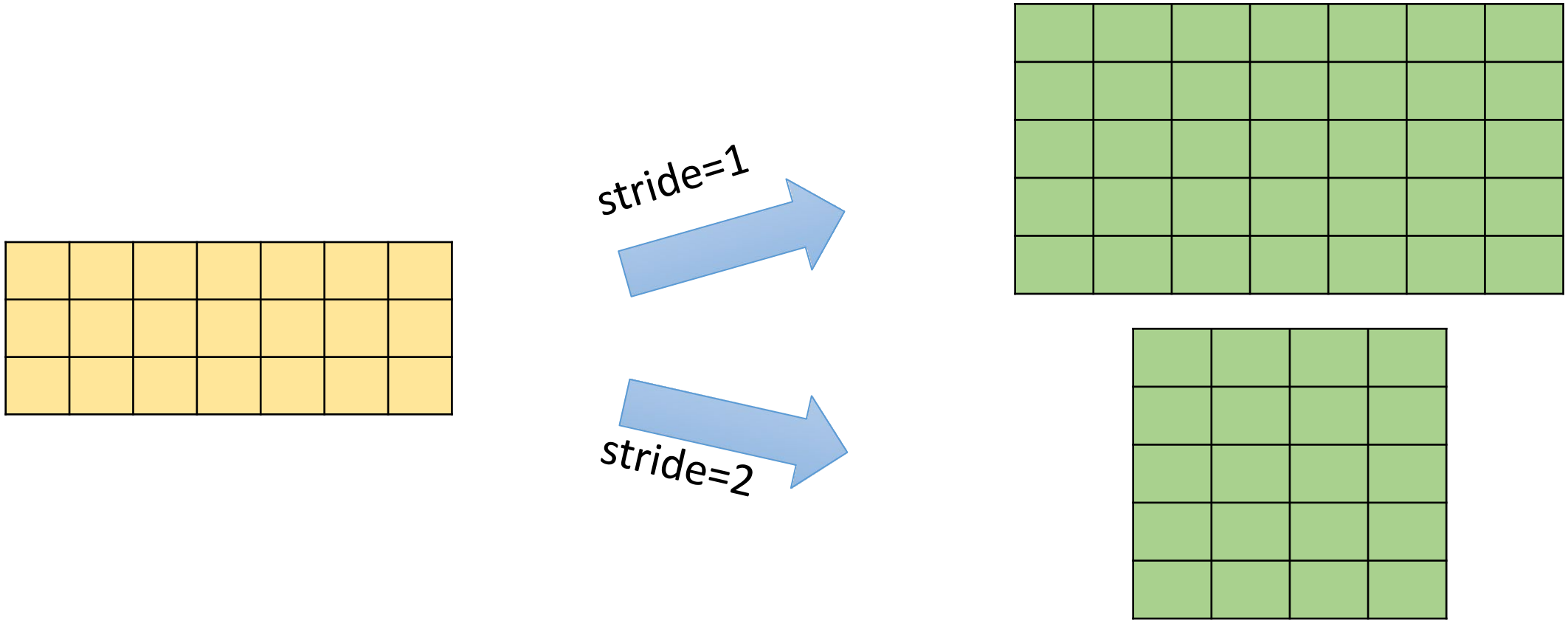
change input from (B = batchsize, T = max time length, 27 = features) to (B, 27, T)

- Transpose/Permute:
  - PyTorch conv1d expects tensors of shape (N, C, L)
    i.e. (batch size, in channels, length)
  - Permuting the input aligns the feature dim with C:
    (B, T, 27) → (B, 27, T)
- Apply convolution (B, 27, T) → (B, 64, T)
- Transpose/Permute: (B, 64, T) → (B, T, 64)
- Pack and pass to sequence model

Assuming *batch_first = True*
(You may also have it as (T, B, 27)

Embedding Layer: Conv1d Layers

If stride > 1, we effectively reduce the time steps



Embedding Layer: Conv1d Layers

- Stride > 1 reduces computation for LSTM and training is faster.
- However, too much reduction in time steps will lead to loss of information (we don’t recommend downsampling more than 4x)

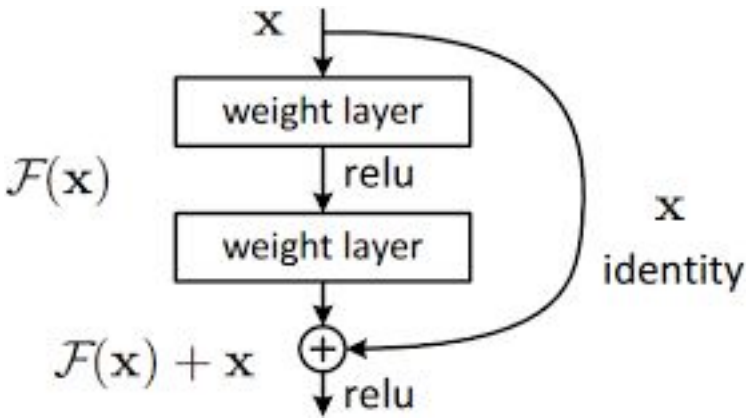
Embedding Layer: Conv1d Layers

- Stride > 1 reduces computation for LSTM and training is faster.
- However, too much reduction in time steps will lead to loss of information (we don't recommend downsampling more than 4x)

- **Note: Stride > 1 alters number of time steps. You need to change lengths_x accordingly**
  - Use convolution formula ***(X – K + 2*P)//S*** (or)
  - Clamp lengths to length of embedding (torch function)

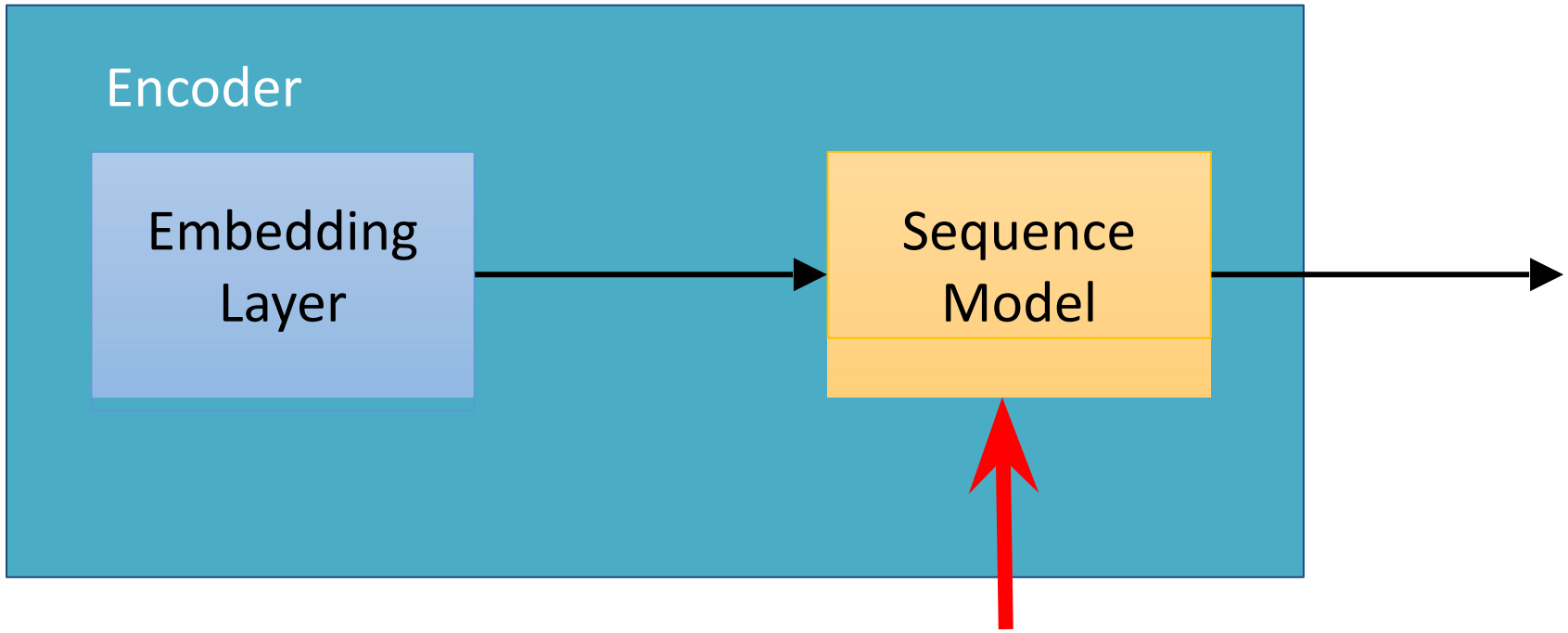
Embedding Layer: Conv1d Layers

- You can try convolution layers based on residual blocks
- Hint: Remember HW2P2!



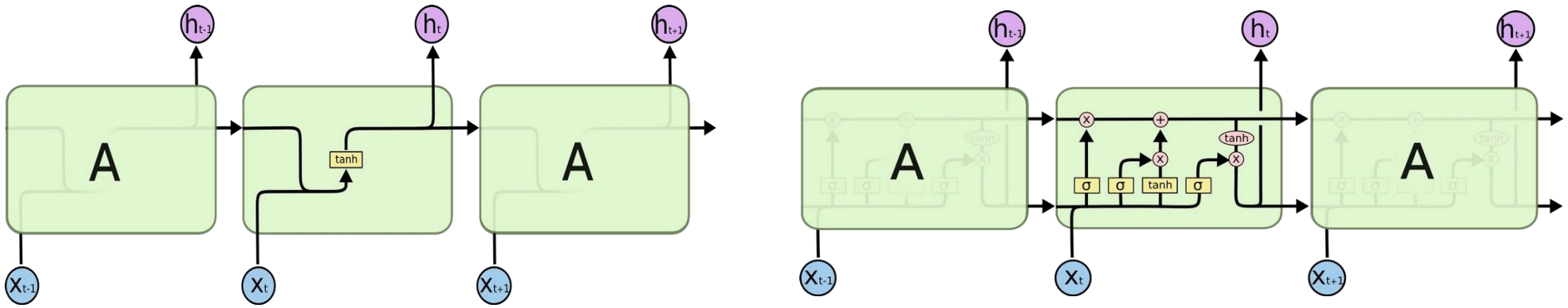
https://www.cv-foundation.org/openaccess/content_cvpr_2016/papers/He_Deep_Residual_Learning_CVPR_2016_paper.pdf

Sequence Model



Sequence Model

- Can use RNN, GRU, **LSTM** (recommended) from *torch.nn*
  - Expects packed_padded sequence method (check documentation)



http://colah.github.io/posts/2015-08-Understanding- LSTMs/

Sequence Model

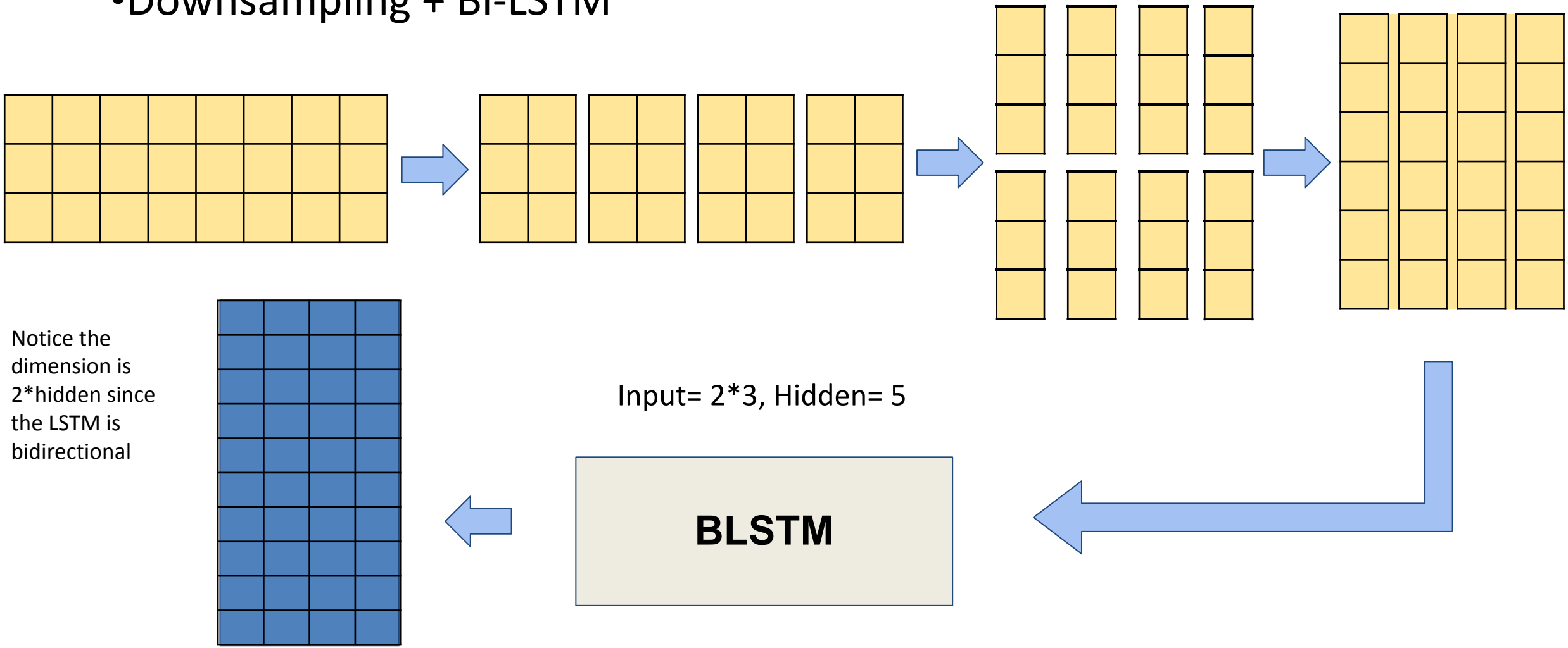
- Important parameters/hyper parameters in *nn.LSTM()*
  - *input_size (27 or embedding_size)*
  - *hidden_dim*
  - *num_layers*
  - *dropout (An aside on dropout: Don't use nn.dropout(p), use nn.LSTM(dropout=p) instead)*
  - *bidirectional*
  - Note: when *bidirection = True*, LSTM outputs a shape of *hidden_dim* in the forward direction and *hidden_dim* in the backward direction (in total, *2*hidden_dim*)

pBLSTM

- **p**yramidal **B**i-directional **LSTM.** Described in the Listen-Attend-Spell paper
- The pBLSTM is a variant of Bi-LSTMs that downsamples sequences by a factor of 2 by **concatenating adjacent pairs of inputs** before running a conventional Bi-LSTM on the reduced-length sequence
- This can be implemented using reshape

Pyramidal Bi-LSTM (pBLSTM)

- Downsampling + Bi-LSTM

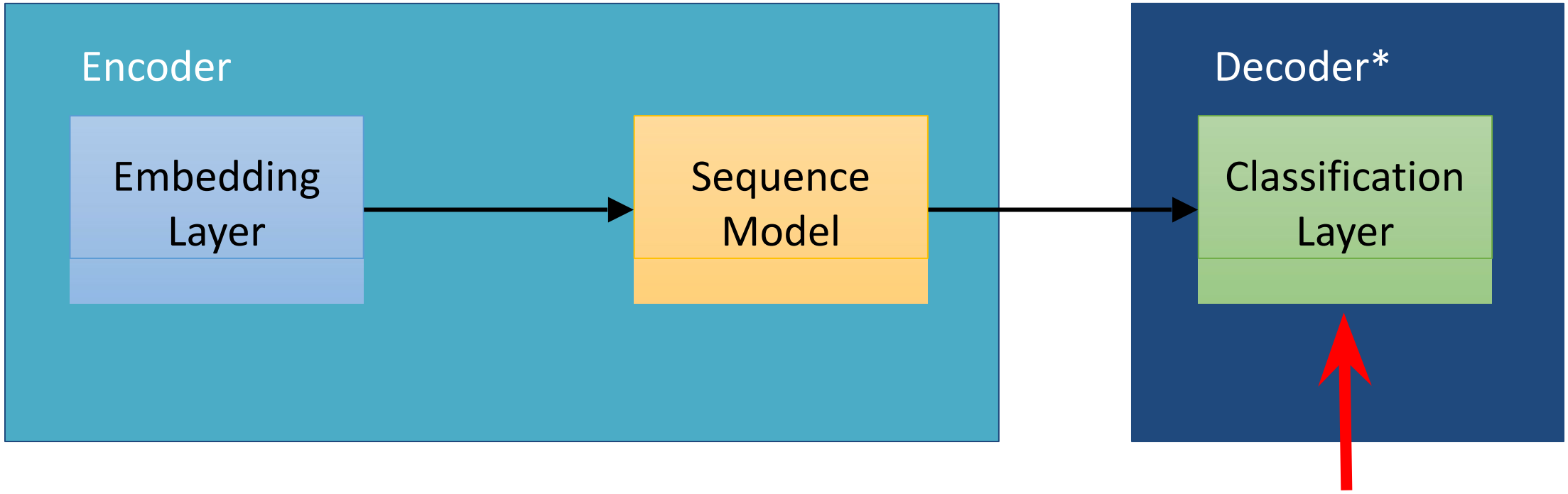


pBLSTM - pseudocode

Listing 1 pBLSTM

```
# X = (batchsize, length, dim) is a minibatch of input sequences, possibly from a previous layer
# Assuming dataloader ensures that all input sequences in the batch are the same length
function O = pBLSTM(X, LSTMwidth, Params)
    # Reshape inputs to have half the length, but twice the dimensionality
    X_downsampled = reshape(X,B,L/2,2*D)
    output = BiLSTM(X_downsampled, LSTMwidth, Params)
    return output
end
```

Classification Layer



Classification Layer

- Same as HW1P2 - just an MLP
- Output from the sequence model goes to the classification layer
- Variations
  - Deeper
  - Wider
  - Different activations
  - Dropout

Hyperparameters and Regularization

- Cepstral Normalization:

$$X \rightarrow (X - \text{mean})/\text{std}$$

- Different weight initialization (for Conv and Linear layers)
- Weight decay with optimizer

Hyperparameters and Regularization

- Scheduler is very important
  - ReduceLRonPlateau (Most of our ablation)
    - Lev distance might start to oscillate at lower values
  - Step LR
- Optimizer
  - AdamW, Adam
- Learning Rate - start with a small learning rate (1e-3)

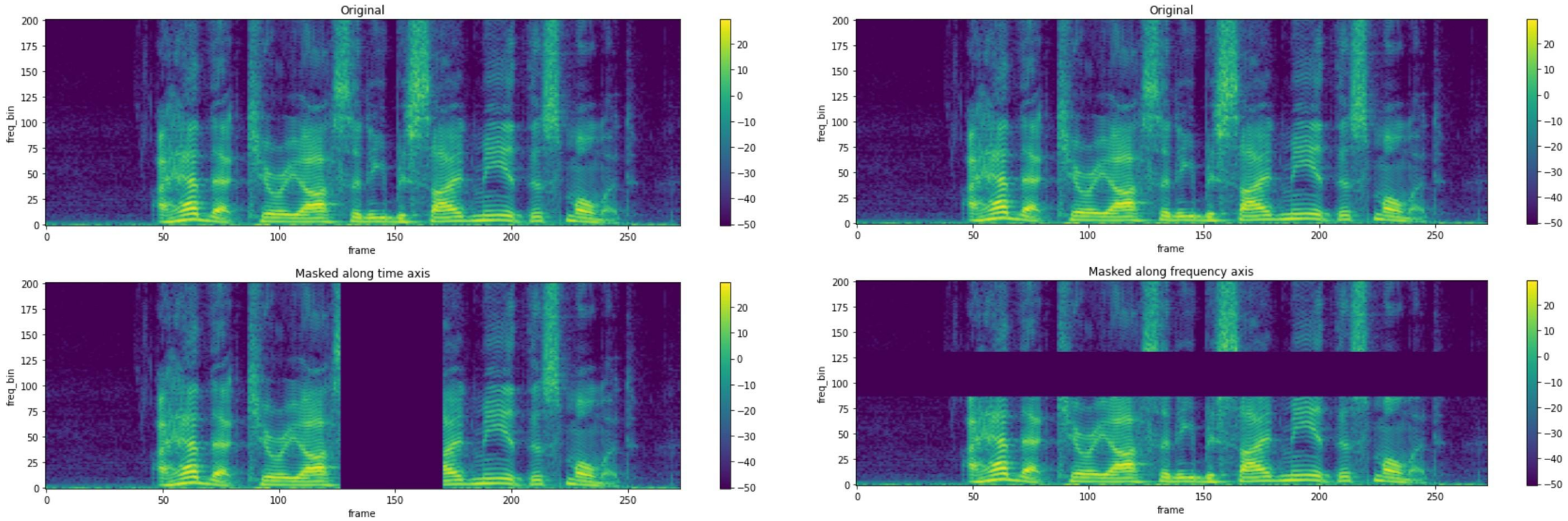
Hyperparameters and Regularization

- Dropout is key
  - Can use dropout in all the 3 layers: Embedding, Sequence model and classification
  - You can also start with a small dropout rate and increase after the model gets trained

- Locked Dropout for LSTM layer
  - Locked Dropout can be used to apply the same dropout mask to every time step
  - You can refer to PyTorch NLP's implementation of locked dropout here
  - Pay attention to whether modules adhere to batch first format or not

Hyperparameters and Regularization

- Torch Audio Transforms [docs]
  - Time Masking (vertical)
  - Frequency Masking (horizontal)



Hyperparameters and Regularization

- Beam width
  - Higher beam width may give better results (advisable to keep test beam width below 50 for computation purposes)
  - Sometimes bw = 1 (greedy search) also gives good results
  - Tip: Don’t use a high beam width while validating in each epoch (time per epoch will be higher)

Final Tips

- Make sure to split work within your study groups
- Don’t forget to also attend lab 9 (CTC and Beam Search) on **March 13th**! The slides go into a good bit of detail on ctcloss, beam search, among other things!

Good Luck🤗